SUPERMARKET SALES ANALYSIS

An Excel Based Data Analysis Project

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Course: Bachelor of Science in Information Technology

Institution: Dedan Kimathi University of Technology

Tools used: Microsoft Excel

Project Type: Data Analysis and Visualization

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1.Introduction

1.1 Background

Sales data provides valuable information about a business's performance, customer purchasing behavior, and operational efficiency. Analyzing this data helps businesses understand

sales trend, identify important customers, and make informed decisions.

This project focuses on analyzing Supermarket sales data using Microsoft Excel. The dataset contains 70 sales records with information such us Order Numbers, Order Dates, Customer Names, Ship Dates, Retail Prices, Order quantities, taxes and total sales.

1.2 Problem Statement

Businesses may collect sales records without fully understanding the patterns and insights contained in them. Without proper analysis, it can be difficult to identify high performing customers, compare sales across periods, and evaluate shipping performance.

Tis project addresses this challenge by organizing, analyzing, and visualizing supermarket sales data using Microsoft Excel.

1.3 Purpose of the Project

The purpose of this project is to transform raw supermarket sales data into meaningful information through data inspection, data cleaning, calculations, Pivot tables, charts, and an interactive-style dashboard.

2. Project Objectives

2.1 General Objective

To analyze Supermarket sales data using Microsoft Excel in order to identify the sales trends, customer purchasing patterns, and shipping performance.

2.2 Specific Objectives

The specific objectives of the project are:

1. To inspect and clean the supermarket sales dataset to improve data quality.
2. To calculate key sales metrics, including total revenue, total orders, average order value, and total quantities sold.
3. To analyze sales performance by month, customer, and order size.
4. To create PivotTables, charts, and dashboard to present the finding clearly.
5. To provide recommendations based on the results of the analysis.

3. Tools and Methodology

3.1 Tools Used

The following tools were used in the project:

- **Microsoft Excel**: Used for data inspection, data cleaning, calculations, PivotTables, charts and dashboard creation.
- **Microsoft Word**: Used to prepare the project report and document the findings.

3.2 Dataset Description

The dataset contains 70 sales records with information about orders, customers, prices, quantities, taxes, and shipping dates. Additional fields including shipping days, order size category, and order month were created to support further analysis.

3.3 Methodology

The project followed the following steps:

1. **Data Inspection**: The dataset structure, column headings and data were examined to understand the information available.
2. **Data Cleaning:** Unnecessary columns were removed and formulas were corrected to improve the working dataset.
6. **Data Calculation:** Excel formulas were used to calculate sales metrics which included total revenue, total orders, average order value, and total quantities sold.
3. **Data Analysis**: Formulas and PivotTables were used to analyze monthly sales, customer sales, order sizes, order quantities and shipping performance.
4. **Data Visualization**: Charts were created to present the analysis results in a clear and understandable format.
5. **Dashboard Development**: Key Performance Indicator's (KPI's) and charts were combined into a sales dashboard to provide an overview of the supermarket's performance.
6. **Reporting**: This report was prepared to document the projects objectives, methods, findings and recommendations.

4. Finding and Visualizations

4.1 Key Sales Metrics

The analysis used key performance indicators (KPIs) to summarize supermarket sales performance. These included total revenue, total orders, average order value, total quantities sold, highest sale and lowest sale. These sales provide an overview of the sales recorded in the dataset.

The results were as follows:

Total Revenue: $10,855.15

Total Orders: 70

Total Quantity Sold: 168

Average Order Value: $155.07

Highest Sale: $329.97

Lowest Sale: $32.99

SALES METRICS	VALUE
Total Revenue	$10,855.15
Total Orders	70
Total Quantity Sold	168
Average Order Value	$155.07
Highest Sale	$329.97
Lowest Sale	$32.99

Figure 1: Key Sales Performance Indicators

The key sales metrics summarize the supermarket's overall sales performance. They show total revenue, the number of orders, the quantity sold, the average order value, and the highest and lowest sales recorded in the dataset.

4.2 Monthly Sales Analysis

A PivotTable and a chart were created to analyze sales revenue by month. This visualization helps compare sales performance across the months represented in the dataset and identify the differences in monthly revenue.

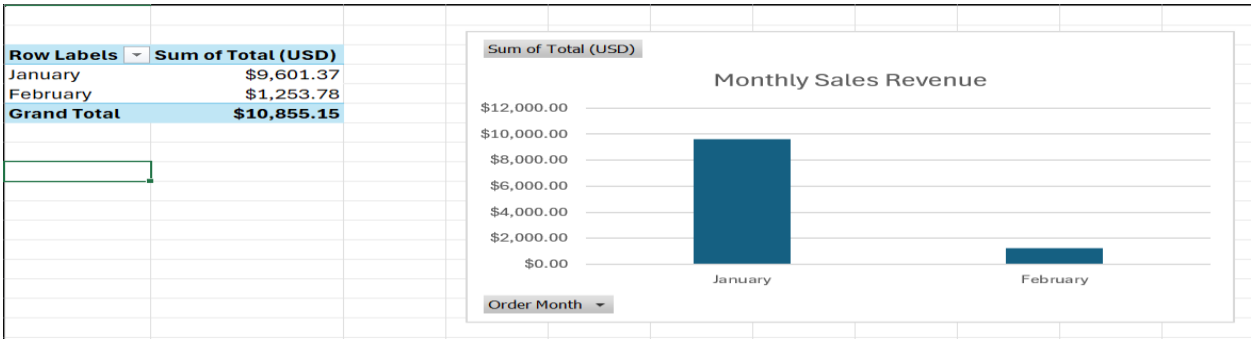
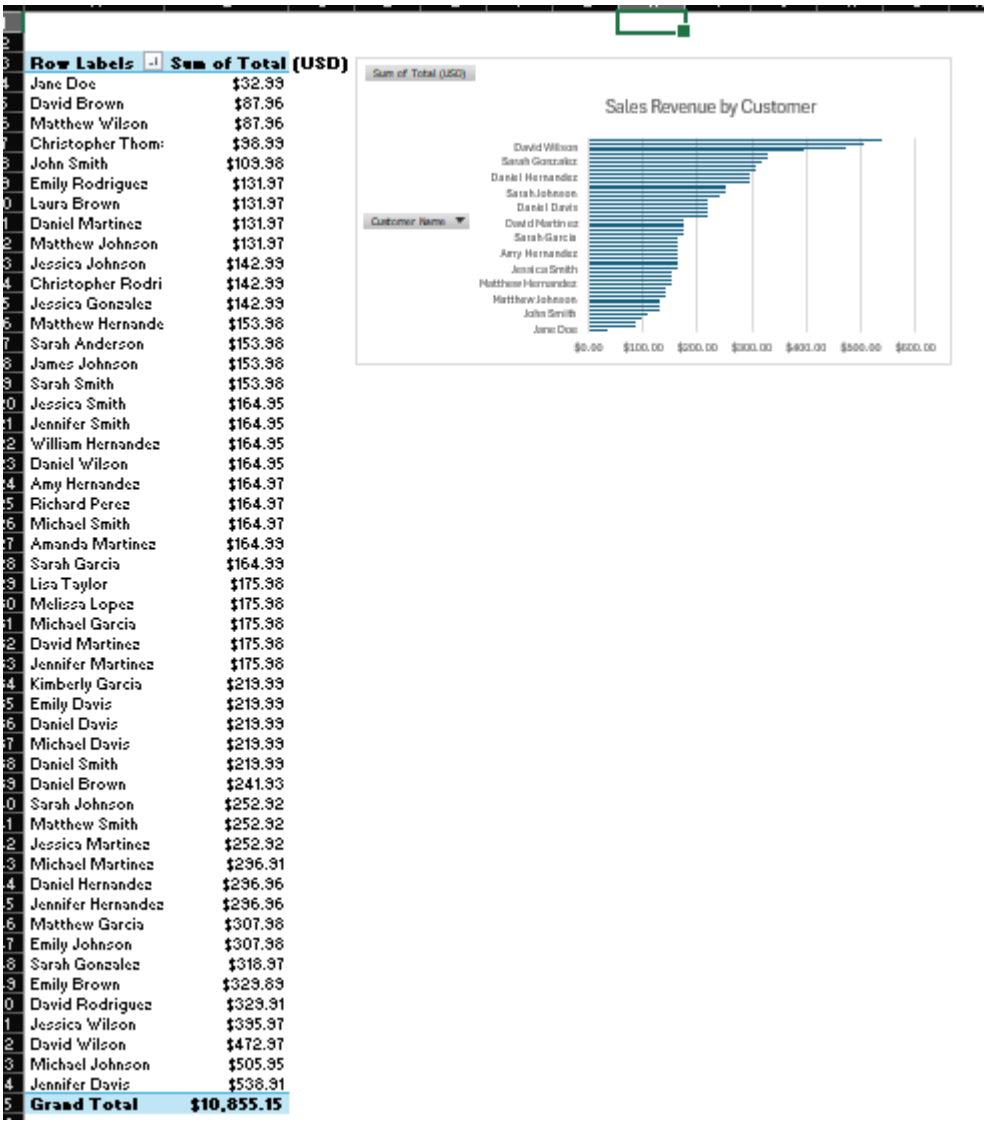


Figure 2: Monthly Sales Revenue Analysis

The analysis compares revenue generated by different customers, helping identify the highest-value customers.

4.3 Customer Sales Analysis

Customer sales were analyzed using a PivotTable and chart showing total sales revenue associated with each customer. This helps identify customers who contributed more revenue during the period covered by the dataset.



Row Labels	Sum of Total (USD)
Jane Doe	$32.99
David Brown	$87.96
Matthew Wilson	$87.96
Christopher Thom	$98.99
John Smith	$109.98
Emily Rodrigues	$131.97
Laura Brown	$131.97
Daniel Martinez	$131.97
Matthew Johnson	$131.97
Jessica Johnson	$142.99
Christopher Rodri	$142.99
Jessica Gonzalez	$142.99
Matthew Hernande	$153.98
Sarah Anderson	$153.98
James Johnson	$153.98
Sarah Smith	$153.98
Jessica Smith	$164.95
Jennifer Smith	$164.95
William Hernandez	$164.95
Daniel Wilson	$164.95
Amy Hernandez	$164.97
Richard Perez	$164.97
Michael Smith	$164.97
Amanda Martinez	$164.99
Sarah Garcia	$164.99
Lisa Taylor	$175.98
Melissa Lopez	$175.98
Michael Garcia	$175.98
David Martinez	$175.98
Jennifer Martinez	$175.98
Kimberly Garcia	$219.99
Emily Davis	$219.99
Daniel Davis	$219.99
Michael Davis	$219.99
Daniel Smith	$219.99
Daniel Brown	$241.93
Sarah Johnson	$252.92
Matthew Smith	$252.92
Jessica Martinez	$252.92
Michael Martinez	$296.91
Daniel Hernandez	$296.96
Jennifer Hernandez	$296.96
Matthew Garcia	$307.98
Emily Johnson	$307.98
Sarah Gonzalez	$318.97
Emily Brown	$329.89
David Rodriguez	$329.91
Jessica Wilson	$395.97
David Wilson	$472.97
Michael Johnson	$505.95
Jennifer Davis	$538.91
Grand Total	**$10,855.15**

Figure 3: Sales Revenue By Customer

The analysis compares customer contributions to total sales revenue.

4.4 Order Size and Quantity Analysis

The analysis compared the revenue from large and small orders and examined revenue by order quantity. The analysis recorded $1,154.62 from Large orders and $9,700.54 from small orders. These visualizations help show how order sizes and order quantities relate to sales revenue.

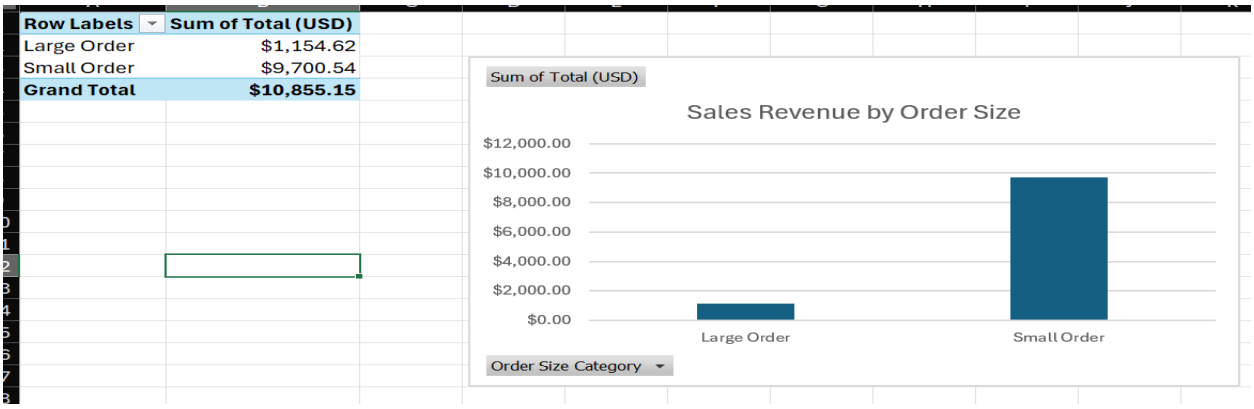


Figure 4: Sales Revenue By Order Size

The analysis compares revenue generated by large and small orders.

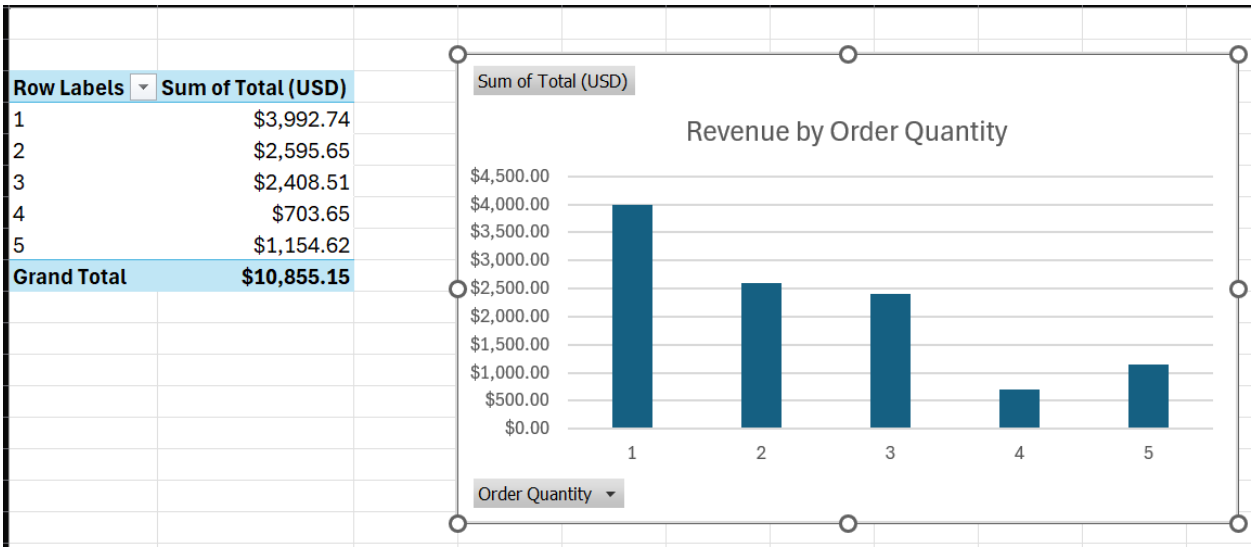


Figure 5: Sales Revenue by Order Quantity

The analysis shows how revenue varies according to the quantity purchased per order.

4.5 Shipping *Performance* Analysis

Shipping Performance was examined by calculating the number of days between order dates and shipping dates. A chart was created to show the number of orders associated with different shipping durations.

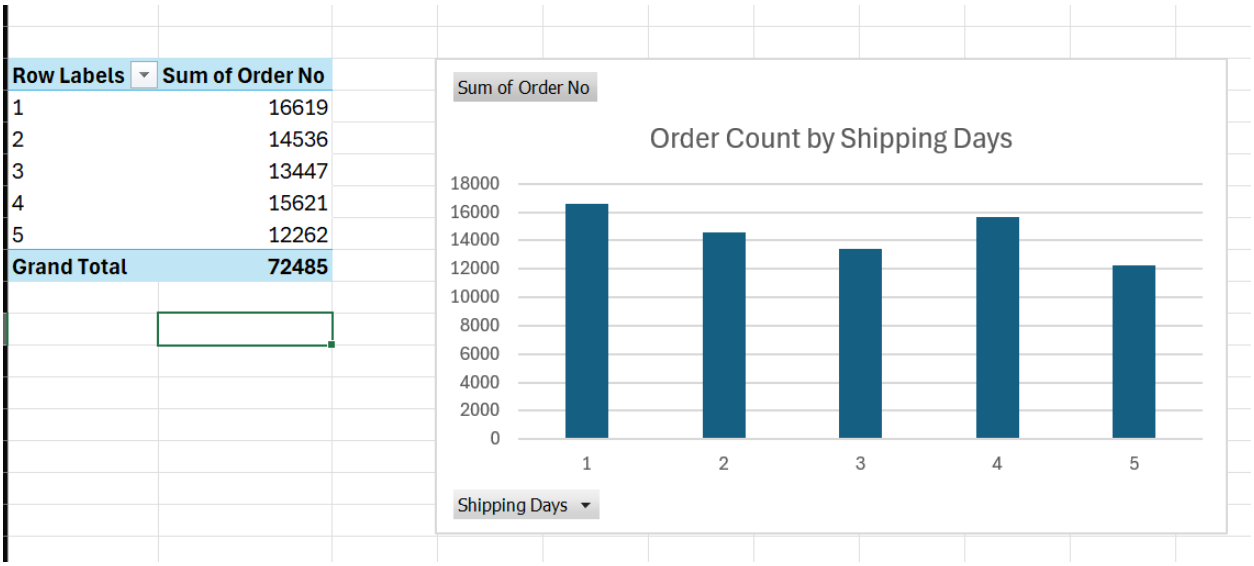


Figure 6: Order Count by Shipping Days

The analysis shows how long orders take to ship, helping assess delivery performance.

4.6 Sales Dashboard

A dashboard was developed to bring together the key sales metrics and charts in one worksheet. It provides a convenient overview of sales performance and makes it easier to compare different aspects of the dataset.

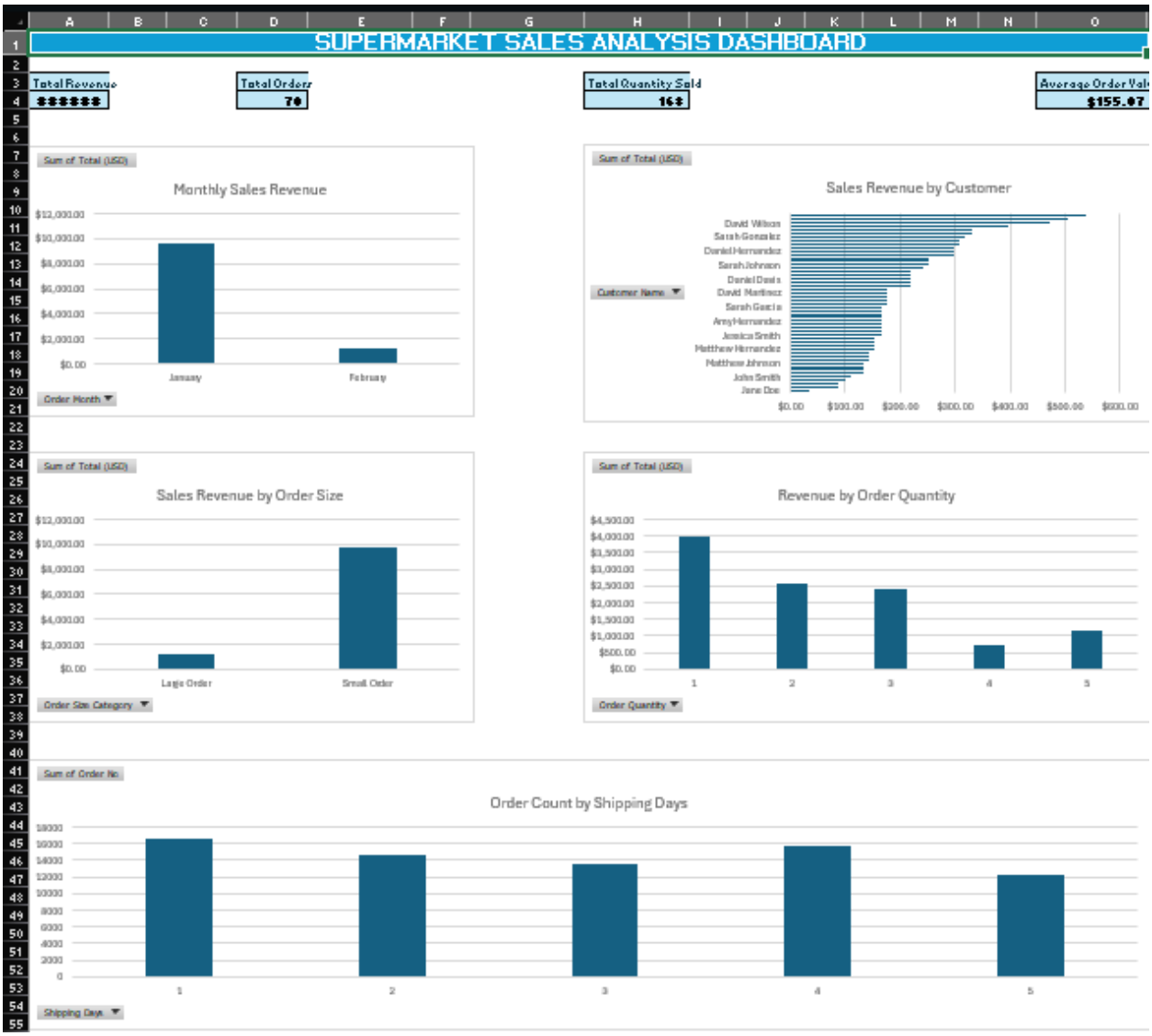


Figure 7: Supermarket Sales Analysis Dashboard

The dashboard brings together key sales metrics and charts to provide a clear overview of supermarket performance.

5. Recommendations and Conclusion

5.1 Recommendations

Based on the analysis performed, the following recommendations are proposed:

1. **Monitor Sales Trend:** The Supermarket should regularly review monthly sales to understand changes in revenue and support better planning.
2. **Understand Customer Contributions:** Customer sales analysis should be used to identify valuable customers and guide customer retention strategies.
3. **Review Order Sizes:** The supermarket should examine the contribution of large and small orders when planning sales.
4. **Monitor Shipping Activities:** Shipping durations should be reviewed regularly to identify opportunities to improve order delivery efficiency.
5. **Use Data for Decision Making:** The dashboard should be updated with new sales records to support ongoing performance monitoring and informed business decisions.

5.2 Conclusion

This project demonstrated how Microsoft Excel can be used to transform supermarket sales data into useful business information. Data inspection, cleaning, formulas, PivotTables, charts, and a dashboard were applied to analyze sales performance, customer contributions, order sizes, quantities, and shipping durations.

The project also provided practical experience in spreadsheet-based data analysis and visualization. The resulting dashboard presents key sales indicators and analysis results in an organized format, supporting clearer interpretation of the available data and more informed decision-making.